

Topic: AI Search Lab — 8-Puzzle Solver & Tic-Tac-Toe with AI

Comparative Analysis

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Structural Comparison:

The search space of an 8-puzzle is similar to the game tree of tic-tac-toe in that both pursue an optimal route to a goal state. In the 8-puzzle, the optimal solution is the shortest path from the start state to the goal state. In tic-tac-toe, the optimal solution is the move that produces the best possible outcome assuming both players play optimally. It differs in that tic-tac-toe has two players competing against each other and both players are trying to win individually, while the puzzle solver is a single-agent problem.

Algorithm Fit:

As mentioned in the structural comparison, A* is not applicable to Tic-Tac-Toe because it is an algorithm for a single agent. Minimax does not apply to 8-puzzle because it is an adversarial search player. The 8-puzzle game does not have two opposing players so there is no assignable mini or max.

Empirical Comparison - Module A:

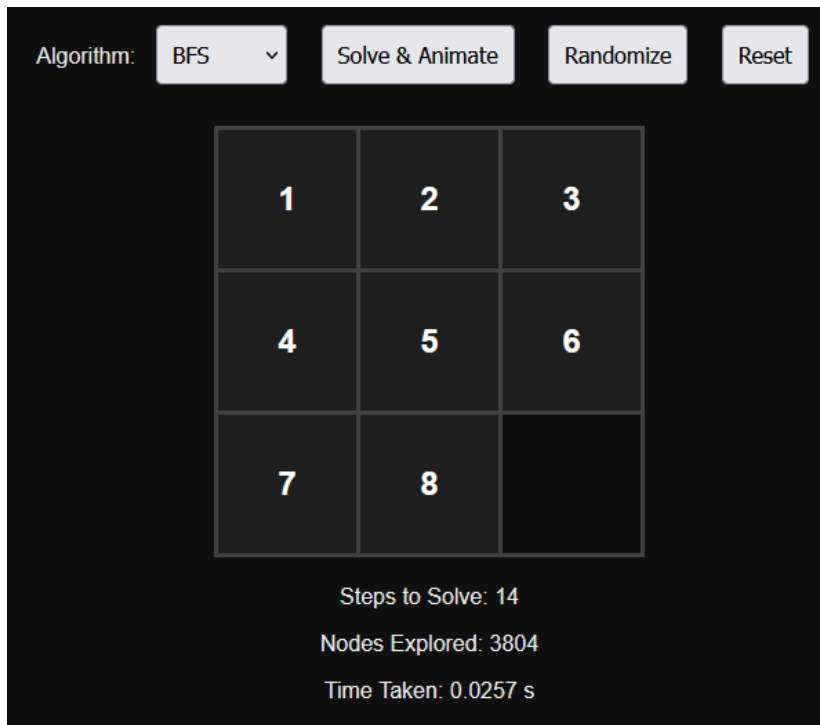


Figure 1. BFS Search Results

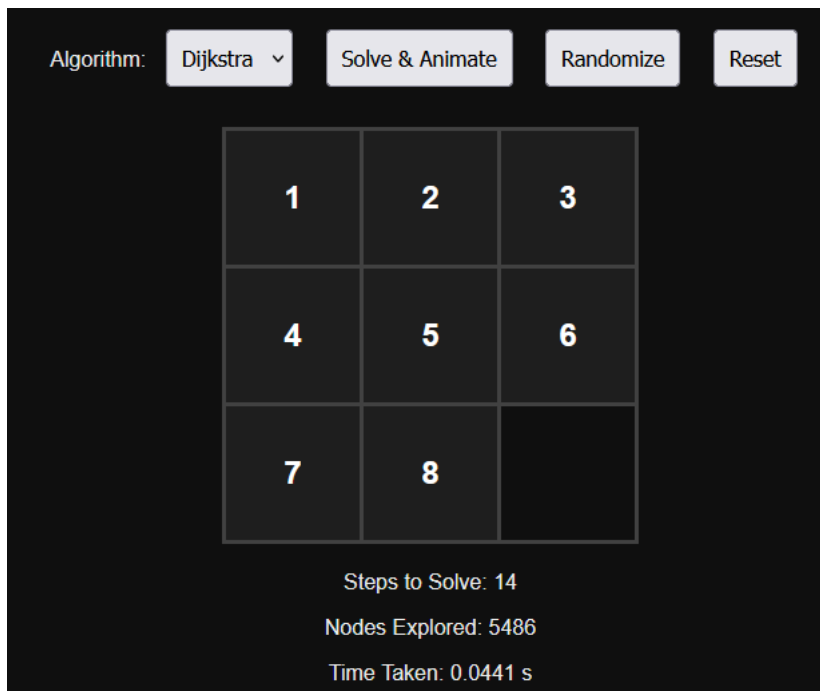


Figure 2. Dijkstra's Search Results

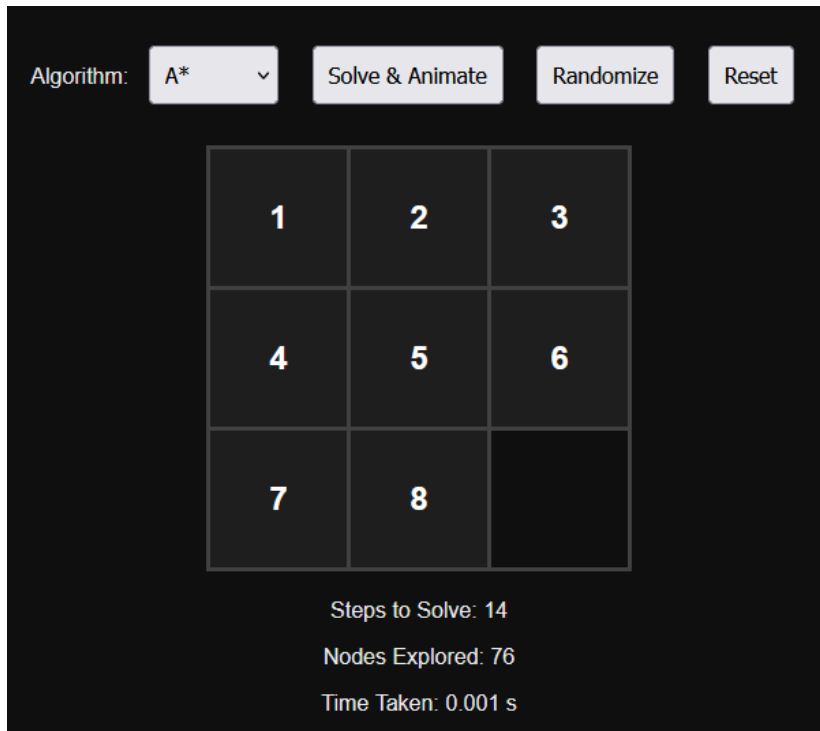


Figure 3. A* Search

Standardized test indicates massive improvement in performance with A* search. The amount of nodes explored dropped significantly which attributes to the value of heuristic information. The Manhattan distance is useful in traversing a search space that will lead to configurations with the fewest moves needed. All algorithms are able to find the shortest path but runtime is by far the shortest with A*.

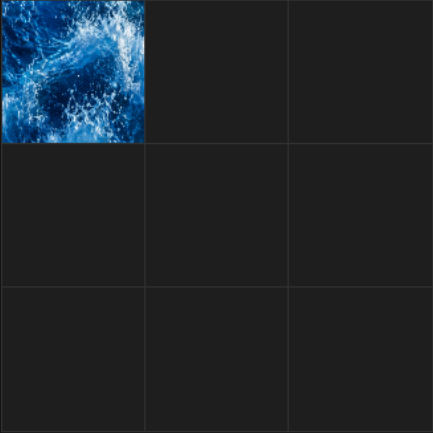
It was surprising to see Dijkstra's algorithm explore the most nodes. It may be because it is taking more time to visit nodes with fewer steps instead of level-by-level. However, including both a count of steps and the heuristic as seen with A* search is extremely effective.

Empirical Comparison - Module B:

Tic Tac Toe AI

[Go to 8 Puzzle Solver](#)

Your Turn (X)



Mode:

AI Algorithm:

Moves: 1

Nodes Explored: 549945

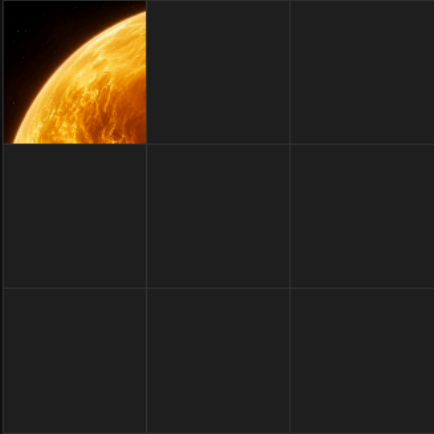
Decision Time: 1.4525

Pruning Rate: --

Tic Tac Toe AI

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Your Turn (X)



Mode:

AI Algorithm:

Moves: 1

Nodes Explored: 28035

Decision Time: 1.6179

Pruning Rate: 94.9%

Alpha-Beta Pruning algorithm reports a 94.9% pruning rate. The search space has been pruned dramatically for efficiency. It is considered that the decision time is longer because it is performing minimax first to be able to calculate the pruning rate.

Heuristic Justification

Manhattan Distance is the heuristic used for the A* search. It is admissible because it calculates the total of distances each tile is from the goal position. The puzzle cannot be solved in fewer moves than the total Manhattan distance suggests. Thus, the true solution length is never overestimated.